ytauto – complete upload script (Python)

Run once: python auth_youtube.py • Upload: ytauto.upload.upload(video, metadata, kind, thumbnail)

auth_youtube.py

```
"""One-time YouTube login. Opens a browser; sign in with the CHANNEL's Google account and click Allow.

Creates token.json next to this file. Copy token.json to the server together with the code.
"""
from google_auth_oauthlib.flow import InstalledAppFlow

from ytauto import config
from ytauto.upload import SCOPES

flow = InstalledAppFlow.from_client_secrets_file(str(config.CLIENT_SECRET_FILE), SCOPES)
creds = flow.run_local_server(port=0, access_type="offline", prompt="consent")
config.TOKEN_FILE.write_text(creds.to_json(), encoding="utf-8")
print(f"Saved {config.TOKEN_FILE}")
```

ytauto/upload.py

```
"""Upload to YouTube with the Data API v3 using the saved OAuth token."""
from google.auth.transport.requests import Request
from google.oauth2.credentials import Credentials
from googleapiclient.discovery import build
from googleapiclient.http import MediaFileUpload

from . import config

SCOPES = [
    "https://www.googleapis.com/auth/youtube.upload",
    "https://www.googleapis.com/auth/youtube.readonly",
]


def _youtube():
    creds = Credentials.from_authorized_user_file(str(config.TOKEN_FILE), SCOPES)
    if not creds.valid:
        creds.refresh(Request())
        config.TOKEN_FILE.write_text(creds.to_json(), encoding="utf-8")
    return build("youtube", "v3", credentials=creds, cache_discovery=False)


def upload(video_path, script, kind, thumbnail_path=None):
    hashtags = " ".join(script["hashtags"])
    title = script["title"]
    if kind == "short" and "#shorts" not in title.lower():
        title = f"{title} #Shorts"

    body = {
        "snippet": {
            "title": title[:100],
            "description": f"{script['description']}\n\n{hashtags}"[:5000],
            "tags": script["tags"],
            "categoryId": config.YOUTUBE_CATEGORY_ID,
            "defaultLanguage": "hi",
            "defaultAudioLanguage": "hi",
        },
        "status": {
            "privacyStatus": config.UPLOAD_PRIVACY,
            "selfDeclaredMadeForKids": False,
            # Honest AI disclosure: synthetic voice narration.
            "containsSyntheticMedia": True,
        },
    }

    youtube = _youtube()
    request = youtube.videos().insert(
        part="snippet,status",
```

```
        body=body,
        media_body=MediaFileUpload(str(video_path), chunksize=8 << 20, resumable=True),
    )
    response = None
    while response is None:
        _, response = request.next_chunk()
    video_id = response["id"]

    if thumbnail_path:
        youtube.thumbnails().set(videoId=video_id, media_body=MediaFileUpload(str(thumbnail_path))).execute()
    return video_id
```

ytauto/config.py

```
import os
from pathlib import Path

from dotenv import load_dotenv

ROOT = Path(__file__).resolve().parent.parent
load_dotenv(ROOT / ".env")

OUTPUT_DIR = ROOT / "output"
DATA_DIR = ROOT / "data"
MUSIC_DIR = ROOT / "assets" / "music"
CLIENT_SECRET_FILE = ROOT / "client_secret.json"
TOKEN_FILE = ROOT / "token.json"

MODEL = os.getenv("ANTHROPIC_MODEL", "claude-opus-5-5")
PEXELS_API_KEY = os.getenv("PEXELS_API_KEY", "")

CHANNEL_NAME = os.getenv("CHANNEL_NAME", "Kya Aap Jaante Hain")
CHANNEL_NICHE = os.getenv(
    "CHANNEL_NICHE",
    "Hindi facts and explainers about technology, AI, science, space, money and everyday life",
)

TTS_VOICE = os.getenv("TTS_VOICE", "hi-IN-MadhurNeural")
TTS_RATE = os.getenv("TTS_RATE", "+8%")

UPLOAD_PRIVACY = os.getenv("UPLOAD_PRIVACY", "private")
YOUTUBE_CATEGORY_ID = os.getenv("YOUTUBE_CATEGORY_ID", "27")

SUBTITLE_FONT = os.getenv("SUBTITLE_FONT", "Noto Sans Devanagari")
FPS = 30

# Per-format settings. Hindi narration runs at roughly 130-150 words per minute.
FORMATS = {
    "short": {
        "width": 1080,
        "height": 1920,
        "orientation": "portrait",
        "words": "110 to 140 words (about 45-55 seconds)",
        "scenes": "6 to 9",
        "font_size": 84,
        "words_per_caption": 3,
    },
    "long": {
        "width": 1920,
        "height": 1080,
        "orientation": "landscape",
        "words": "900 to 1100 words (about 7-8 minutes)",
        "scenes": "35 to 45",
        "font_size": 58,
        "words_per_caption": 6,
    },
}
```